EXP NO: 6

Case Study on Food Delivery Applications – Positive Test Cases

Objective

To identify and analyze the requirements of Food Delivery Applications and prepare positive test scenarios and test cases to verify the proper functioning of the application.

Features to be Tested

1. User Login and Authentication.
2. Restaurant Search and Selection.
3. Food Menu and Item Selection.
4. Add Food Items to Cart.
5. Apply Offers and Discounts.
6. Delivery Address Management.
7. Online Payment.
8. Order Confirmation and Tracking.
9. Order History.
10. Profile and Review Management.

Bug ID and Bug Name

Bug ID	Bug Name
FD 001	Valid User Login
FD 002	Restaurant Search
FD 003	Food Menu Display
FD 004	Add Food Items to Cart
FD 005	Cart Quantity Update
FD 006	Apply Valid Coupon
FD 007	Delivery Address Selection
FD 008	Successful Payment

FD 009 Order Confirmation

FD 010 Order Tracking

FD 011 Order History

FD 012 Rating and Review

BUG REPORT

Bug Id: FD 001

Bug Description: Valid User Login

Steps to reproduce:

1. Open the food delivery application.
2. Enter the valid username and password.
3. Click on the Login button.

Expected Result: Home screen should be displayed.

Actual Result: Home screen is displayed successfully.

Status: Pass.

Bug Id: FD 002

Bug Description: Restaurant Search

Steps to reproduce:

1. Log in to the food delivery application.
2. Enter a valid delivery location.
3. Click on Search Restaurants.
4. Select an available restaurant.

Expected Result: Available restaurants should be displayed based on the delivery location.

Actual Result: Restaurants are displayed successfully.

Status: Pass.

Bug Id: FD 003

Bug Description: Food Menu Display

Steps to reproduce:

1. Open the food delivery application.
2. Select a restaurant.
3. Click on the Menu section.
4. View the available food items.

Expected Result: Food items, prices and descriptions should be displayed.

Actual Result: Food menu is displayed successfully.

Status: Pass.

Bug Id: FD 004

Bug Description: Add Food Items to Cart

Steps to reproduce:

1. Open a restaurant.
2. Select an available food item.
3. Click on Add to Cart.
4. Open the cart.

Expected Result: Selected food item should be added to the cart.

Actual Result: Food item is added successfully to the cart.

Status: Pass.

Bug Id: FD 005

Bug Description: Cart Quantity Update

Steps to reproduce:

1. Add food items to the cart.
2. Open the cart.
3. Increase or decrease the quantity of selected food items.
4. Check the updated total amount.

Expected Result: Updated quantity and total amount should be displayed correctly.

Actual Result: Quantity and total amount are updated successfully.

Status: Pass.

Bug Id: FD 006

Bug Description: Apply Valid Coupon

Steps to reproduce:

1. Add food items to the cart.
2. Open the coupon section.
3. Enter a valid coupon code.
4. Click on Apply.

Expected Result: Valid coupon should be applied and the discount should be displayed.

Actual Result: Coupon is applied successfully.

Status: Pass.

Bug Id: FD 007

Bug Description: Delivery Address Selection

Steps to reproduce:

1. Open the cart.
2. Click on Proceed to Checkout.
3. Enter a valid delivery address.
4. Click on Save Address.

Expected Result: Delivery address should be saved successfully.

Actual Result: Delivery address is saved successfully.

Status: Pass.

Bug Id: FD 008

Bug Description: Successful Payment

Steps to reproduce:

1. Proceed to checkout.
2. Select a valid payment method.
3. Enter valid payment details.

4. Click on Pay Now.

Expected Result: Payment should be completed successfully.

Actual Result: Payment is completed successfully.

Status: Pass.

Bug Id: FD 009

Bug Description: Order Confirmation

Steps to reproduce:

1. Complete the payment process.
2. Open the order confirmation page.
3. Check the order details.

Expected Result: Order ID, ordered food items and total amount should be displayed.

Actual Result: Order confirmation details are displayed successfully.

Status: Pass.

Bug Id: FD 010

Bug Description: Order Tracking

Steps to reproduce:

1. Log in to the application.
2. Click on My Orders.
3. Select the current order.
4. Check the order tracking status.

Expected Result: Current order status should be displayed.

Actual Result: Order tracking details are displayed successfully.

Status: Pass.

Bug Id: FD 011

Bug Description: Order History

Steps to reproduce:

1. Open the Profile section.

2. Click on Order History.
3. View the previous orders.

Expected Result: Previous order details should be displayed.

Actual Result: Order history is displayed successfully.

Status: Pass.

Bug Id: FD 012

Bug Description: Rating and Review

Steps to reproduce:

1. Open a completed order.
2. Click on Rate and Review.
3. Select a valid rating.
4. Enter a review.
5. Click on Submit.

Expected Result: Rating and review should be submitted successfully.

Actual Result: Rating and review are submitted successfully.

Status: Pass.

Result

Thus, the requirements of the Food Delivery Application were identified and analyzed. The positive test scenarios and test cases were prepared to verify the major functionalities of the application, including user login, restaurant search, food selection, cart management, payment, order tracking and review management.

EXP NO: 7

Case Study on Food Delivery Applications – Negative Test Cases

Objective

To identify and analyze the requirements of Food Delivery Applications and prepare negative test scenarios and test cases to verify how the application handles invalid inputs, incorrect data and unsuccessful user actions.

Features to be Tested

1. Home Page and User Login.
2. Food Order and Favourite Items.
3. Restaurant Awards.
4. Zomato UPI and Zomato Gold.
5. Events and Event Tickets.
6. Dining Transactions and Rewards.
7. Money Gift Cards.
8. Language Selection and Feedback.
9. Profile Management and Reviews.
10. User Ratings.

Negative Test Scenarios

1. Home Page: Verify the application behavior when an incorrect email address or password is entered.
2. Food Order: Verify whether incorrect food items are displayed when accessing favourite orders.
3. Restaurant Awards: Verify whether fake or incorrect restaurant names are displayed in the restaurant awards section.
4. Zomato UPI: Verify whether incorrect UPI transactions or insufficient balance messages are handled properly.
5. Events: Verify whether incorrect event tickets are displayed.
6. Dining Transactions: Verify whether money is incorrectly debited from the bank account during dining reward bookings.
7. Money Gift Cards: Verify whether unauthorized users can access or purchase gift cards.
8. Language Selection: Verify whether the application fails to send feedback after changing the language.
9. Profile Management: Verify whether users are unable to add reviews or upload profile photos.
10. User Ratings: Verify whether incorrect or missing ratings are displayed.

Test Environment

- Approach: Manual Approach
- Test Data: Username / Password
- Test Environment: Operating System / Browser
- Training Needed: Not Required
- Risk: No

BUG ID AND BUG NAME

1. FD 001 – Invalid Login Credentials
2. FD 002 – Incorrect Food Items in Favourite Orders
3. FD 003 – Fake Restaurant Names
4. FD 004 – Incorrect UPI Transaction
5. FD 005 – Invalid Event Tickets
6. FD 006 – Incorrect Dining Transaction
7. FD 007 – Unauthorized Gift Card Access
8. FD 008 – Feedback Submission Failure
9. FD 009 – Profile Review Failure
10. FD 010 – Incorrect User Rating

BUG REPORT

Bug Id: FD 001

Bug Description: Invalid Login Credentials

Steps to reproduce:

1. Open the food delivery website.
2. Enter an incorrect email address or password.
3. Click on the Login button.
4. Check the login response.

Expected Result: Invalid email or password message should be displayed.

Actual Result: User should not be allowed to log in using incorrect credentials.

Status: Pass.

Bug Id: FD 002

Bug Description: Incorrect Food Items in Favourite Orders

Steps to reproduce:

1. Open the food delivery application.
2. Click on Food Order.
3. Select Favourite Orders.
4. Check the displayed food items.

Expected Result: Only the food items added to favourites should be displayed.

Actual Result: Incorrect food items are displayed in the favourite orders section.

Status: Pass.

Bug Id: FD 003

Bug Description: Fake Restaurant Names

Steps to reproduce:

1. Open the food delivery application.
2. Click on Restaurant Awards.
3. Select Winning Restaurants.
4. Check whether the restaurant names are correct.

Expected Result: Only genuine winning restaurant names should be displayed.

Actual Result: Fake or incorrect restaurant names are displayed.

Status: Pass.

Bug Id: FD 004

Bug Description: Incorrect UPI Transaction

Steps to reproduce:

1. Open the food delivery application.
2. Click on Zomato UPI.
3. Check the UPI activity.
4. Verify the transaction details and wallet balance.

Expected Result: Incorrect transactions should not be accepted, and insufficient balance messages should be displayed when applicable.

Actual Result: Incorrect transaction acceptance or insufficient balance message may appear.

Status: Pass.

Bug Id: FD 005

Bug Description: Invalid Event Tickets

Steps to reproduce:

1. Open the food delivery application.
2. Click on Events.
3. Select Your Events.
4. Check the event tickets displayed on the screen.

Expected Result: Only valid event tickets should be displayed.

Actual Result: Wrong event tickets are displayed on the screen.

Status: Pass.

Bug Id: FD 006

Bug Description: Incorrect Dining Transaction

Steps to reproduce:

1. Open the food delivery application.
2. Click on Dining.
3. Select Dining Rewards.
4. Book a dining slot.
5. Check the bank transaction details.

Expected Result: Money should be debited only after a successful booking, and the correct dining reward should be applied.

Actual Result: Money may be debited incorrectly from the bank account without a successful booking.

Status: Pass.

Bug Id: FD 007

Bug Description: Unauthorized Gift Card Access

Steps to reproduce:

1. Open the food delivery application.
2. Click on Money.
3. Select Buy Gift Cards.
4. Try to access the gift card section without valid authorization.

Expected Result: Unauthorized users should not be allowed to access or purchase gift cards.

Actual Result: Unauthorized access to the gift card section may occur.

Status: Pass.

Bug Id: FD 008

Bug Description: Feedback Submission Failure

Steps to reproduce:

1. Open the application.
2. Select the language option.
3. Choose a different language.
4. Open the feedback section.
5. Enter feedback and click Submit.

Expected Result: Feedback should be submitted successfully.

Actual Result: Application is unable to send the feedback.

Status: Pass.

Bug Id: FD 009

Bug Description: Profile Review Failure

Steps to reproduce:

1. Open the Profile section.
2. Select Add Review.
3. Enter the review details.
4. Try to upload a profile photo.

5. Click on Submit.

Expected Result: Review and profile photo should be submitted successfully.

Actual Result: Application is unable to submit the review or upload the photo.

Status: Pass.

Bug Id: FD 010

Bug Description: Incorrect User Rating

Steps to reproduce:

1. Open the food delivery application.
2. Navigate to Your Ratings.
3. Select a previously rated food item.
4. Check whether the submitted rating is displayed correctly.

Expected Result: User rating should be displayed correctly.

Actual Result: Incorrect or missing user rating is displayed.

Status: Pass.

Result

Thus, the requirements of the Food Delivery Application were identified and analyzed. Negative test scenarios and test cases were prepared to verify the application's behavior when invalid credentials, incorrect information, unauthorized access and unsuccessful transactions are encountered.

EXP NO: 8

Case Study on Banking Applications – Positive Test Cases

Objective

To identify and analyze the requirements of Banking Applications and prepare positive test scenarios and test cases to verify the proper functioning of the banking application.

Features to be Tested

1. User Registration.
2. User Login.

3. Account Management.
4. Fund Transfer.
5. Bill Payment.
6. Account Security.
7. Customer Support.
8. Mobile Banking.
9. Transaction Notifications.
10. Integration with External Services.

Positive Test Scenarios

1. User Registration: Verify that a user can successfully register using valid information and a unique username.
2. User Login: Verify that a registered user can log in using valid credentials.
3. Account Management: Verify that users can create bank accounts, view account details and update personal information.
4. Fund Transfer: Verify that users can successfully transfer money between bank accounts.
5. Bill Payment: Verify that users can pay electricity, water and telephone bills successfully.
6. Account Security: Verify that users can update passwords and enable two-factor authentication.
7. Customer Support: Verify that users can contact customer support through chat, email and phone.
8. Mobile Banking: Verify that the mobile application provides a user-friendly experience and supports banking functionalities.
9. Transaction Notifications: Verify that users receive notifications for successful transactions and account changes.
10. Integration with External Services: Verify that the banking application integrates successfully with external payment gateways and services.

Test Environment

- Approach: Manual Approach
- Test Data: Username and Password

- Test Environment: Laptop / Mobile Phone
- Training Needed: No
- Risk: No

BUG ID AND BUG NAME

1. BA 001 – User Registration
2. BA 002 – User Login
3. BA 003 – Account Management
4. BA 004 – Fund Transfer
5. BA 005 – Bill Payment
6. BA 006 – Account Security
7. BA 007 – Customer Support
8. BA 008 – Mobile Banking
9. BA 009 – Transaction Notifications
10. BA 010 – Integration with External Services

BUG REPORT

Bug Id: BA 001

Bug Description: User Registration

Steps to reproduce:

1. Open the banking application.
2. Click on the Registration option.
3. Enter a valid name, email address and password.
4. Enter a unique username.
5. Click on Register.

Expected Result: User should be successfully registered with valid information.

Actual Result: User is registered successfully.

Status: Pass.

Bug Id: BA 002

Bug Description: User Login

Steps to reproduce:

1. Open the banking application.
2. Enter the registered email address and password.
3. Click on Login.
4. Check whether the user is logged in successfully.

Expected Result: User should be able to log in using valid credentials.

Actual Result: User is logged in successfully.

Status: Pass.

Bug Id: BA 003

Bug Description: Account Management

Steps to reproduce:

1. Log in to the banking application.
2. Select the Account Management option.
3. Create a new bank account.
4. View account details, including balance and transaction history.
5. Update personal information.

Expected Result: User should be able to create an account, view account details and update information successfully.

Actual Result: Account details are displayed and updated successfully.

Status: Pass.

Bug Id: BA 004

Bug Description: Fund Transfer

Steps to reproduce:

1. Log in to the banking application.
2. Select the Fund Transfer option.
3. Enter the receiver's valid account number.

4. Enter the transfer amount.
5. Confirm the transaction.

Expected Result: Money should be transferred successfully, and the amount should be accurately reflected in both sender and receiver accounts.

Actual Result: Fund transfer is completed successfully.

Status: Pass.

Bug Id: BA 005

Bug Description: Bill Payment

Steps to reproduce:

1. Log in to the banking application.
2. Select the Bill Payment option.
3. Choose electricity, water or telephone bill.
4. Enter valid bill details.
5. Confirm the payment.

Expected Result: Bill payment should be completed successfully, and the transaction should be reflected in the user's transaction history and biller's record.

Actual Result: Bill payment is completed successfully.

Status: Pass.

Bug Id: BA 006

Bug Description: Account Security

Steps to reproduce:

1. Log in to the banking application.
2. Open the Account Security section.
3. Select Change Password.
4. Enter the current password and a valid new password.
5. Confirm the new password.
6. Enable two-factor authentication.

Expected Result: New password should be updated successfully and two-factor authentication should be enabled.

Actual Result: Password is updated and two-factor authentication is enabled successfully.

Status: Pass.

Bug Id: BA 007

Bug Description: Customer Support

Steps to reproduce:

1. Log in to the banking application.
2. Open the Customer Support section.
3. Select chat, email or phone support.
4. Submit a valid support request.
5. Check the response.

Expected Result: Customer support request should be submitted and an appropriate response should be received.

Actual Result: Customer support request is submitted successfully.

Status: Pass.

Bug Id: BA 008

Bug Description: Mobile Banking

Steps to reproduce:

1. Install and open the mobile banking application.
2. Log in using valid credentials.
3. Navigate through the available banking features.
4. Check the application on supported mobile devices.

Expected Result: Mobile banking application should provide a user-friendly experience and all major banking functionalities should work properly.

Actual Result: Mobile banking application works successfully.

Status: Pass.

Bug Id: BA 009

Bug Description: Transaction Notifications

Steps to reproduce:

1. Log in to the banking application.
2. Perform a successful fund transfer.
3. Check the transaction notification.
4. Verify the transaction details.
5. Check notification settings.

Expected Result: User should receive accurate notifications for successful transactions and account changes.

Actual Result: Transaction notifications are received successfully with accurate details.

Status: Pass.

Bug Id: BA 010

Bug Description: Integration with External Services

Steps to reproduce:

1. Open the banking application.
2. Select an external payment service.
3. Enter valid payment details.
4. Complete the payment.
5. Check whether the transaction information is updated.

Expected Result: Banking application should integrate smoothly with external services and exchange transaction data successfully.

Actual Result: External service integration is completed successfully.

Status: Pass.

Result

Thus, the requirements of the Banking Application were identified and analyzed. Positive test scenarios and test cases were prepared to verify the major functionalities, including user registration, login, account management, fund transfer, bill payment, account security, customer support and transaction notifications.

EXP NO: 9

CASE STUDY ON BANKING APPLICATIONS – NEGATIVE TEST CASES

Objective:

To identify and analyze the requirements of Banking Applications and to prepare negative test scenarios and test cases to verify how the application handles invalid inputs, incorrect credentials, and unsuccessful transactions.

Features to be Tested

1. User Registration
2. User Login
3. Account Management
4. Bill Payment
5. Account Security

Negative Test Scenarios

1. Verify that the application does not allow registration with invalid email addresses, weak passwords, or missing mandatory fields.
2. Verify that the application displays an appropriate error message when a user enters incorrect login credentials.
3. Verify that the application does not allow users to access account details using invalid account information.
4. Verify that the application displays an error when a bill payment fails or the bill details are invalid.
5. Verify that the application rejects passwords that do not meet security requirements.

Test Environment

- Approach: Manual Approach
- Test Data: Invalid username, password, email, and account details
- Environment: Laptop and Mobile Phone
- Training Needed: NO
- Risk: NO

Bug ID and Bug Name

- BA 001 – Invalid User Registration
- BA 002 – Invalid User Login
- BA 003 – Invalid Account Management
- BA 004 – Bill Payment Failure
- BA 005 – Invalid Password Security

BUG REPORTS

Bug ID: BA 001

Bug Description: User Registration – Application should reject invalid registration details.

Steps to Reproduce:

Step 1 – Verify that the application does not allow users to register with an invalid email address.

Step 2 – Verify that mandatory fields cannot be left empty.

Step 3 – Verify that the application rejects duplicate usernames and weak passwords.

Expected Result: The application should display appropriate error messages for invalid registration details and should not create the account.

Actual Result: The application rejects invalid registration details and displays appropriate error messages.

Status: Pass

Bug ID: BA 002

Bug Description: User Login – Application should reject incorrect login credentials.

Steps to Reproduce:

Step 1 – Enter an invalid username or email address.

Step 2 – Enter an incorrect password and click the Login button.

Step 3 – Verify that an appropriate error message is displayed.

Step 4 – Verify that the user is not allowed to access the account with incorrect credentials.

Expected Result: The application should reject incorrect credentials and display an appropriate error message.

Actual Result: The application rejects the login attempt and displays an error message indicating that the credentials are incorrect.

Status: Pass

Bug ID: BA 003

Bug Description: Account Management – Application should not allow access using invalid account information.

Steps to Reproduce:

Step 1 – Enter an invalid account number while accessing account details.

Step 2 – Verify that the application does not display account details for an invalid account.

Step 3 – Try to update account information using invalid details.

Expected Result: The application should reject invalid account information and display an appropriate error message.

Actual Result: The application does not display account details and rejects the invalid information.

Status: Pass

Bug ID: BA 004

Bug Description: Bill Payment – Application should handle unsuccessful bill payments correctly.

Steps to Reproduce:

Step 1 – Enter an invalid bill reference number or incorrect bill details.

Step 2 – Try to make a payment using invalid information.

Step 3 – Verify that the application displays an appropriate error message.

Step 4 – Verify that the failed payment is not recorded as a successful transaction.

Expected Result: The application should reject the invalid bill payment and display an appropriate error message. The unsuccessful transaction should not appear as a successful payment in the transaction history.

Actual Result: The application rejects the invalid payment and does not record it as a successful transaction.

Status: Pass

Bug ID: BA 005

Bug Description: Account Security – Application should reject passwords that do not satisfy security requirements.

Steps to Reproduce:

Step 1 – Enter the current password incorrectly while attempting to change the password.

Step 2 – Enter a new password that does not meet the required length and complexity.

Step 3 – Verify that the application rejects the invalid password.

Step 4 – Verify that the application displays an appropriate error message.

Expected Result: The application should reject incorrect current passwords and weak new passwords. It should display an appropriate error message without updating the password.

Actual Result: The application rejects the incorrect password and does not allow the user to set a weak password.

Status: Pass

Result: Thus, negative test scenarios and test cases for the Banking Application were prepared to verify how the application handles invalid registration details, incorrect login credentials, invalid account information, unsuccessful bill payments, and weak passwords.

EXP NO: 10

LIBRARY MANAGEMENT SYSTEM – TEST PLAN

Objective:

To develop a comprehensive test plan for the Library Management System by identifying its major features and preparing test cases to verify the functionality of registration, book searching, borrowing, returning, book addition, report generation, security, and membership renewal.

Test Plan

Test Approach: Manual Approach
Test Environment: Web Browser
Training Needed: NO
Risk: NO

1. User Registration

Feature: User Authentication
Approach: Manual Approach
Test Data: Username and Password
Test Environment: Browser

Training Needed: NO
Risk: NO

Test Cases:

Step 1 – Open the Library Management System website.

Step 2 – Enter a valid username and password.

Step 3 – Click the Register or Login button.

Step 4 – Verify that the user is successfully authenticated and the home page appears.

Expected Result: The user should be able to register or log in successfully.

Status: Pass

2. Searching for a Book

Feature: Book Identification
Approach: Manual Approach
Test Data: Book Categories
Test Environment: Browser
Training Needed: NO
Risk: NO

Test Cases:

Step 1 – On the home page, click the book category.

Step 2 – Select the required category.

Step 3 – Choose a book from the selected category.

Step 4 – Verify that the selected book details are displayed.

Expected Result: The system should display the required book and its details.

Status: Pass

3. Borrowing a Book

Feature: Book Borrowing
Approach: Manual Approach
Test Data: Book Selection
Test Environment: Browser
Training Needed: NO
Risk: NO

Test Cases:

Step 1 – On the home page, select the required book category.

Step 2 – Choose a book from the category.

Step 3 – Select the required book.

Step 4 – Click the Borrow option.

Step 5 – Verify that the book is successfully borrowed.

Expected Result: The system should allow the user to borrow an available book and update its availability status.

Status: Pass

4. Returning a Book

Feature: Book Returning
Approach: Manual Approach
Test Data: Borrowed Book Details
Test Environment: Browser
Training Needed: NO
Risk: NO

Test Cases:

Step 1 – Open the borrowed books section.

Step 2 – Select the book that needs to be returned.

Step 3 – Click the Return option.

Step 4 – Verify that the book is successfully returned.

Expected Result: The system should update the book status and make it available for other users.

Status: Pass

5. Adding a New Book to the Library

Feature: New Book Addition
Approach: Manual Approach
Test Data: Book Availability and Details
Test Environment: Browser
Training Needed: NO
Risk: NO

Test Cases:

Step 1 – Select the category of the book.

Step 2 – Enter the book details and the number of copies.

Step 3 – Click the Add Book option.

Step 4 – Verify that the new book is added to the library.

Expected Result: The system should successfully add the new book and update the library records.

Status: Pass

6. Number of Books in the Library

Feature: Number of Books
Approach: Manual Approach
Test Data: Number of Books in Each Category
Test Environment: Browser
Training Needed: NO
Risk: NO

Test Cases:

Step 1 – Open the book categories section.

Step 2 – Add or view the books available in each category.

Step 3 – Verify that the total number of books is displayed.

Expected Result: The system should correctly identify and display the total number of books available in the library.

Status: Pass

7. Generating a Report of Borrowed Books

Feature: Generalized Report of Borrowing
Approach: Manual Approach
Test Data: Number of Borrowed Books and Date
Test Environment: Browser
Training Needed: NO
Risk: NO

Test Cases:

Step 1 – Open the borrowed books section.

Step 2 – Select the date for which the report needs to be generated.

Step 3 – Click the Generate Report option.

Step 4 – Verify that the report is generated successfully.

Expected Result: The system should generate a report containing the borrowed book details for the selected date.

Status: Pass

8. Book ID Number

Feature: Identification of the Book
Approach: Manual Approach
Test Data: Book ID Number
Test Environment: Browser
Training Needed: NO
Risk: NO

Test Cases:

Step 1 – Select the required book.

Step 2 – Verify that each book contains a unique ID number.

Step 3 – Click or search using the Book ID number.

Step 4 – Verify that the correct book details are displayed.

Expected Result: The system should identify the book using its unique Book ID number.

Status: Pass

9. Security

Feature: Authentication and Access Control
Approach: Manual Approach
Test Data: User Login Credentials and Access Control
Test Environment: Browser
Training Needed: NO
Risk: NO

Test Cases:

Step 1 – Open the Security option on the home page.

Step 2 – Select the required security options.

Step 3 – Configure the security settings according to the user's requirements.

Step 4 – Verify that unauthorized users cannot access restricted information.

Expected Result: The system should maintain user security and protect library information from unauthorized access.

Status: Pass

10. Renewal Payment / Membership

Feature: Membership Renewal and Payment
Approach: Manual Approach
Test Data: Username and Password
Test Environment: Browser
Training Needed: NO
Risk: NO

Test Cases:

Step 1 – Click the Payment option on the home page.

Step 2 – If membership renewal is required, select the renewal option.

Step 3 – Enter the required membership details.

Step 4 – Complete the payment process.

Step 5 – Verify that the membership is renewed successfully.

Expected Result: The system should process the payment successfully and update the membership validity.

Status: Pass

Result:

Thus, a test plan for the Library Management System was prepared by identifying its important features and defining the testing approach, test data, test environment, and test cases to verify the functionality of the application.

EXP NO: 11

CASE STUDY ON RAILWAY RESERVATION SYSTEM – POSITIVE TEST CASES

Objective:

To identify and analyze the requirements of a Railway Reservation System and prepare positive test scenarios and test cases to verify the successful functioning of its major features.

Test Plan

Test Approach: Manual Approach
Test Environment: Laptop and Mobile Phone
Training Needed: NO
Risk: NO

1. User Registration

Feature: Name, Email, Mobile Number
Approach: Manual
Test Data: Valid Information, 10-digit Number
Test Environment: Laptop, Phone
Training Needed: NO
Risk: NO

Test Case Scenarios:

Step 1 – Verify that the user can successfully register by entering all valid mandatory fields.

Step 2 – Verify that the system accepts a valid email address and mobile number.

Expected Result: The user should be successfully registered with all valid information.

Status: Pass

2. User Login

Feature: Username, Password
Approach: Manual
Test Data: Valid Credentials
Test Environment: Laptop, Phone
Training Needed: NO
Risk: NO

Test Case Scenarios:

Step 1 – Verify that a registered user can log in to the railway reservation portal successfully.

Step 2 – Verify that the home page is displayed after successful login.

Expected Result: The system should authenticate the registered user and allow access to the portal.

Status: Pass

3. Train Search

Feature: Source, Destination
Approach: Manual
Test Data: Chennai, Bangalore
Test Environment: Laptop, Phone
Training Needed: NO
Risk: NO

Test Case Scenarios:

Step 1 – Enter the source and destination stations.

Step 2 – Click the Search Trains option.

Step 3 – Verify that the system displays available trains between the two valid stations.

Expected Result: The system should display the available trains between the selected source and destination.

Status: Pass

4. Date Selection

Feature: Journey Date
Approach: Manual
Test Data: Future Date
Test Environment: Laptop, Phone
Training Needed: NO
Risk: NO

Test Case Scenarios:

Step 1 – Open the journey date selection option.

Step 2 – Select a valid future date from the calendar.

Step 3 – Verify that the selected journey date is displayed correctly.

Expected Result: The system should allow the user to select a valid future journey date.

Status: Pass

5. Class Selection

Feature: SL, 3A, 2A, 1A
Approach: Manual
Test Data: 3-Tier AC
Test Environment: Laptop, Phone
Training Needed: NO
Risk: NO

Test Case Scenarios:

Step 1 – Select the required train.

Step 2 – Choose the preferred travel class.

Step 3 – Verify that seat availability is updated correctly based on the selected class.

Expected Result: The system should display the correct availability details for the selected travel class.

Status: Pass

6. Passenger Details

Feature: Name, Age, Gender
Approach: Manual
Test Data: Valid Passenger Information
Test Environment: Laptop, Phone
Training Needed: NO
Risk: NO

Test Case Scenarios:

Step 1 – Enter the passenger's name, age, and gender.

Step 2 – Verify that all passenger details are entered correctly.

Step 3 – Verify that the system accepts valid passenger information.

Expected Result: The system should successfully accept and save the passenger details.

Status: Pass

7. Berth Preference

Feature: Lower, Middle, Upper
Approach: Manual
Test Data: Lower Berth
Test Environment: Laptop, Phone

Training Needed: NO
Risk: NO

Test Case Scenarios:

Step 1 – Select the preferred berth option.

Step 2 – Choose Lower, Middle, or Upper berth.

Step 3 – Verify that the system records the selected berth preference.

Expected Result: The system should record and accept the passenger's berth preference.

Status: Pass

8. Payment Gateway

Feature: UPI, Net Banking
Approach: Manual
Test Data: Valid Payment Apps
Test Environment: Laptop, Phone
Training Needed: NO
Risk: NO

Test Case Scenarios:

Step 1 – Select the payment option.

Step 2 – Choose UPI or Net Banking.

Step 3 – Verify that the user is redirected to the payment gateway.

Step 4 – Complete the transaction successfully.

Expected Result: The system should process the payment successfully and confirm the transaction.

Status: Pass

9. PNR Generation

Feature: Digital Ticket
Approach: Manual
Test Data: Transaction ID
Test Environment: Laptop, Phone
Training Needed: NO
Risk: NO

Test Case Scenarios:

Step 1 – Complete the ticket booking and payment process.

Step 2 – Verify that a unique 10-digit PNR is generated after successful payment.

Step 3 – Verify that the PNR is displayed with the booking details.

Expected Result: The system should generate a unique PNR after successful ticket booking.

Status: Pass

10. Ticket Download

Feature: PDF, SMS
Approach: Manual
Test Data: Confirmed PNR
Test Environment: Laptop, Phone
Training Needed: NO
Risk: NO

Test Case Scenarios:

Step 1 – Open the confirmed ticket details.

Step 2 – Click the Download Ticket option.

Step 3 – Verify that the user can download the ticket in PDF format.

Step 4 – Verify that a confirmation SMS is received.

Expected Result: The system should allow the user to download the ticket in PDF format and receive a confirmation SMS.

Status: Pass

Result:

Thus, positive test scenarios and test cases for the Railway Reservation System were prepared to verify the successful functioning of registration, login, train search, date selection, class selection, passenger details, berth preference, payment gateway, PNR generation, and ticket download.

EXP NO: 12

CASE STUDY ON RAILWAY RESERVATION SYSTEM – NEGATIVE TEST CASES

Objective:

To identify and analyze the requirements of a Railway Reservation System and prepare negative test scenarios and test cases to verify how the application handles invalid inputs, incorrect details, and unsuccessful booking operations.

The test plan below follows the same columns and test-case structure shown in your lab manual.

Test Plan

Test Approach: Manual Approach
Test Environment: Laptop and Mobile Phone
Training Needed: NO
Risk: NO

1. User Registration

- Feature: Email Validation
- Approach: Manual
- Test Data: Invalid Email (without @ symbol)
- Test Environment: Laptop, Phone
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Enter an invalid email address without the @ symbol during registration.

Step 2 – Click the Register button.

Step 3 – Verify that the system displays an appropriate error message.

Expected Result: The system should reject the invalid email address and display a validation error.

Status: Pass

2. User Login

- Feature: Security
- Approach: Manual

- Test Data: Wrong Password
- Test Environment: Laptop, Phone
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Enter a registered username with an incorrect password.

Step 2 – Click the Login button.

Step 3 – Verify that the system denies access and displays an "Invalid Credentials" error message.

Expected Result: The system should not allow the user to log in with an incorrect password.

Status: Pass

3. Train Search

- Feature: Route Logic
- Approach: Manual
- Test Data: Same Station as Source and Destination
- Test Environment: Laptop, Phone
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Enter the same station as both source and destination.

Step 2 – Click the Search Trains option.

Step 3 – Verify that the system displays an error message indicating that the source and destination stations are the same.

Expected Result: The system should not allow train searches when the source and destination stations are identical.

Status: Pass

4. Journey Date

- Feature: Date Validation

- Approach: Manual
- Test Data: Past Date
- Test Environment: Laptop, Phone
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Open the journey date selection option.

Step 2 – Try to select a date that has already passed.

Step 3 – Verify that the system restricts the user from selecting the past date.

Expected Result: The system should not allow users to select a past date for a new journey booking.

Status: Pass

5. Passenger Information

- Feature: Age Constraint
- Approach: Manual
- Test Data: Negative Age (-10)
- Test Environment: Laptop, Phone
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Enter passenger details during the booking process.

Step 2 – Enter a negative age such as -10.

Step 3 – Verify that the system rejects the invalid age and displays an appropriate validation message.

Expected Result: The system should not accept negative or invalid passenger ages.

Status: Pass

6. Passenger Name Field

- Feature: Character Limit

- Approach: Manual
- Test Data: Numbers and Special Symbols
- Test Environment: Laptop, Phone
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Open the passenger details section.

Step 2 – Enter numbers or special symbols in the passenger name field.

Step 3 – Verify that the system rejects invalid characters in the name field.

Expected Result: The system should accept valid passenger names and reject inappropriate numbers or special symbols.

Status: Pass

7. Seat Booking

- Feature: Booking Limit
- Approach: Manual
- Test Data: 10 Passengers
- Test Environment: Laptop, Phone
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Start the ticket booking process.

Step 2 – Try to add more passengers than the permitted booking limit.

Step 3 – Verify that the system does not allow booking beyond the maximum passenger limit.

Expected Result: The system should restrict the user from adding more passengers than the permitted limit per ticket.

Status: Pass

8. Payment Gateway

- Feature: Fund Check

- Approach: Manual
- Test Data: Expired Card
- Test Environment: Laptop, Phone
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Proceed to the payment gateway.

Step 2 – Enter expired card details.

Step 3 – Verify that the system handles the payment failure gracefully.

Step 4 – Verify that a "Transaction Failed" notification is displayed.

Expected Result: The system should reject the expired card and display a transaction failure message without confirming the booking.

Status: Pass

9. PNR Inquiry

- Feature: PNR Length
- Approach: Manual
- Test Data: 5-Digit PNR
- Test Environment: Laptop, Phone
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Open the PNR enquiry section.

Step 2 – Enter a PNR number containing only 5 digits.

Step 3 – Verify that the system displays an "Invalid PNR" message when the entered number does not meet the required length.

Expected Result: The system should reject an invalid or incomplete PNR number and display an appropriate error message.

Status: Pass

10. Ticket Cancellation

- Feature: Cancellation Constraint
- Approach: Manual
- Test Data: Departed Train
- Test Environment: Laptop, Phone
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Open the ticket cancellation section.

Step 2 – Select a ticket for a train that has already departed.

Step 3 – Verify that the cancellation option is disabled or an appropriate error message is displayed.

Expected Result: The system should not allow cancellation of a ticket after the train has departed.

Status: Pass

Result:

Thus, negative test scenarios and test cases for the Railway Reservation System were prepared to verify the application's handling of invalid email addresses, incorrect passwords, invalid routes, past journey dates, incorrect passenger details, booking limits, failed payments, invalid PNR numbers, and cancellation restrictions.

EXP NO: 13

CASE STUDY ON ONLINE EXAMINATION SYSTEM – POSITIVE TEST CASES

Objective:

To identify and analyze the requirements of an Online Examination System and prepare positive test scenarios and test cases to verify the successful functioning of its major features.

Test Plan

Test Approach: Manual Approach
Test Environment: Laptop and Browser

Training Needed: NO
Risk: NO

1. Student Registration

- Feature: Student ID, Email
- Approach: Manual
- Test Data: Valid Student Details
- Test Environment: Laptop, Browser
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Verify that the student can successfully register using a valid institutional email address and student ID.

Step 2 – Verify that all mandatory registration details are accepted.

Expected Result: The system should successfully register the student with valid institutional details.

Status: Pass

2. Student Login

- Feature: Authentication Credentials
- Approach: Manual
- Test Data: Valid Username and Password
- Test Environment: Laptop, Browser
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Verify that the registered student can log in using valid credentials.

Step 2 – Verify that the examination dashboard is displayed after successful login.

Expected Result: The system should authenticate the student and provide access to the examination dashboard.

Status: Pass

3. Profile Verification

- Feature: Photo/ID Upload
- Approach: Manual
- Test Data: JPEG/PNG Image
- Test Environment: Laptop, Webcam
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Verify that the system successfully captures or uploads the student's photograph for proctoring.

Step 2 – Verify that the uploaded photograph is displayed correctly.

Expected Result: The system should successfully capture or upload the student's photograph for identity verification.

Status: Pass

4. Exam Selection

- Feature: Subject List
- Approach: Manual
- Test Data: Assigned Subject
- Test Environment: Laptop, Browser
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Verify that the student can view the list of assigned examinations.

Step 2 – Select the required examination.

Step 3 – Verify that the student can access only the examinations for which they are eligible.

Expected Result: The system should display the assigned examinations and allow the student to select an eligible examination.

Status: Pass

5. Instruction Page

- Feature: Guidelines
- Approach: Manual
- Test Data: Click "Agree"
- Test Environment: Laptop, Browser
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Verify that the student can read the examination instructions.

Step 2 – Click the "Agree" button after reading the guidelines.

Step 3 – Verify that the system allows the student to proceed to the examination.

Expected Result: The system should display the examination guidelines and allow the student to start the exam after accepting them.

Status: Pass

6. Question Navigation

- Feature: Next/Previous
- Approach: Manual
- Test Data: Radio Buttons
- Test Environment: Laptop, Browser
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Verify that the student can move between questions using the Next and Previous buttons.

Step 2 – Verify that the student can select an answer using the radio buttons.

Step 3 – Verify that the student can navigate between questions and select options successfully.

Expected Result: The system should allow students to navigate between questions and select their answers successfully.

Status: Pass

7. Save and Mark

- Feature: Review Later
- Approach: Manual
- Test Data: Flag Feature
- Test Environment: Laptop, Browser
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Verify that the student can mark a question for review.

Step 2 – Verify that the marked question appears highlighted in the question palette.

Step 3 – Verify that the student can move between questions and select options successfully.

Expected Result: The system should allow students to mark questions for review and identify them easily using the question palette.

Status: Pass

8. Timer Function

- Feature: Countdown
- Approach: Manual
- Test Data: System Clock
- Test Environment: Laptop, Browser
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Verify that the examination timer starts correctly when the exam begins.

Step 2 – Verify that the remaining time is displayed accurately.

Step 3 – Verify that the timer continues counting down during the examination.

Expected Result: The system should display the correct remaining examination time and update the countdown accurately.

Status: Pass

9. Auto-Submit

- Feature: Session End
- Approach: Manual
- Test Data: Time Expiry
- Test Environment: Laptop, Browser
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Verify that the examination is automatically submitted when the allotted time is completed.

Step 2 – Verify that the student's saved answers are submitted successfully.

Expected Result: The system should automatically submit the examination when the time duration is completed.

Status: Pass

10. Result Generation

- Feature: Scorecard
- Approach: Manual
- Test Data: Valid Answers
- Test Environment: Laptop, Browser
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Verify that the student can view the result after the evaluation is processed.

Step 2 – Verify that the scorecard displays the correct marks.

Step 3 – Verify that the student can download the scorecard if the option is available.

Expected Result: The system should generate and display the student's scorecard accurately after evaluation.

Status: Pass

Result:

Thus, positive test scenarios and test cases for the Online Examination System were prepared to verify the successful functioning of student registration, login, profile verification, exam selection, instructions, question navigation, save and mark, timer function, auto-submit, and result generation.

EXP NO: 14

CASE STUDY ON ONLINE EXAMINATION SYSTEM – NEGATIVE TEST CASES

Objective:

To identify and analyze the requirements of an Online Examination System and prepare negative test scenarios and test cases to verify how the application handles invalid inputs, unauthorized access, session errors, and unsuccessful examination operations.

The following test cases follow the 10 test items shown in your lab manual.

Test Plan

Test Approach: Manual Approach
Test Environment: Laptop and Browser
Training Needed: NO
Risk: NO

1. User Login

- Feature: Authentication
- Approach: Manual
- Test Data: Invalid Password
- Test Environment: Laptop, Browser
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Enter a valid username with an incorrect password.

Step 2 – Click the Login button.

Step 3 – Verify that the system denies access and displays an "Incorrect Password" error after 3 failed attempts.

Expected Result: The system should reject incorrect login credentials and prevent unauthorized access.

Status: Pass

2. Exam Access

- Feature: Eligibility
- Approach: Manual
- Test Data: Expired Exam Link
- Test Environment: Laptop, Browser
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Try to access the examination using an expired exam link.

Step 2 – Verify that the student cannot start the examination after the scheduled time window has closed.

Expected Result: The system should prevent students from accessing examinations after the permitted time window.

Status: Pass

3. Proctoring Check

- Feature: Webcam Access
- Approach: Manual
- Test Data: Camera Disabled
- Test Environment: Laptop, Browser
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Disable the webcam or deny camera permission in the browser.

Step 2 – Try to start the examination.

Step 3 – Verify that the system prevents the examination from starting if the webcam is not detected or access is blocked.

Expected Result: The system should display an appropriate warning and prevent the examination from starting when webcam access is mandatory.

Status: Pass

4. Tab Switching

- Feature: Security
- Approach: Manual
- Test Data: Alt + Tab
- Test Environment: Laptop, Browser
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Start the online examination.

Step 2 – Press Alt + Tab to switch to another application or browser tab.

Step 3 – Verify that the system displays a warning or automatically submits the examination if the student attempts to switch tabs.

Expected Result: The system should detect tab switching and display a warning or apply the configured examination security restrictions.

Status: Pass

5. Input Validation

- Feature: Answer Validation – Field Symbols
- Approach: Manual
- Test Data: Special Symbols
- Test Environment: Laptop, Browser
- Training Needed: NO

- Risk: NO

Test Case Scenarios:

Step 1 – Open a question that accepts only numeric answers.

Step 2 – Enter alphabets or special symbols in the answer field.

Step 3 – Verify that the system rejects invalid characters.

Expected Result: The system should allow only numeric values in numeric-only answer fields and reject alphabets or special symbols.

Status: Pass

6. Multiple Login

- Feature: Concurrent Session
- Approach: Manual
- Test Data: Same Credentials
- Test Environment: 2 Laptops
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Log in to the examination system using the same user ID on two different devices.

Step 2 – Verify that the system logs out the first session if the same user ID is logged in on another device.

Expected Result: The system should prevent unauthorized simultaneous sessions according to its session management policy.

Status: Pass

7. File Upload

- Feature: Assignment Size
- Approach: Manual
- Test Data: 100 MB File
- Test Environment: Laptop, Browser
- Training Needed: NO

- Risk: NO

Test Case Scenarios:

Step 1 – Open the assignment upload section.

Step 2 – Try to upload a file that exceeds the maximum permitted size, such as a 100 MB file when the limit is 5 MB.

Step 3 – Verify that the system rejects the file and displays an appropriate error message.

Expected Result: The system should reject files that exceed the maximum allowed upload size.

Status: Pass

8. Network Loss

- Feature: Connectivity
- Approach: Manual
- Test Data: Wi-Fi Disconnected
- Test Environment: Laptop, Browser
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Start the online examination.

Step 2 – Disconnect the Wi-Fi connection during the examination.

Step 3 – Verify that the system displays a "No Connection" alert.

Step 4 – Verify that the system handles the interruption without losing previously saved answers.

Expected Result: The system should display a network connection alert and preserve saved examination data during network interruption.

Status: Pass

9. Submit Button

- Feature: Blank Submission
- Approach: Manual
- Test Data: No Questions Selected
- Test Environment: Laptop, Browser

- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Start the online examination.

Step 2 – Leave all questions unanswered.

Step 3 – Click the Submit button.

Step 4 – Verify that the system displays a warning before submitting the examination without answering any questions.

Expected Result: The system should display a confirmation or warning message when the student tries to submit without answering questions.

Status: Pass

10. URL Manipulation

- Feature: Unauthorized Access
- Approach: Manual
- Test Data: Direct Result URL
- Test Environment: Laptop, Browser
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Copy the result page URL from an authorized session.

Step 2 – Log out of the examination system.

Step 3 – Try to access the result page directly using the copied URL.

Step 4 – Verify that the student cannot access the result page without proper authentication.

Expected Result: The system should prevent unauthorized users from accessing examination results through a direct URL.

Status: Pass

Result:

Thus, negative test scenarios and test cases for the Online Examination System were prepared to verify the application's handling of incorrect login credentials, expired exam links, disabled webcams, tab switching, invalid answers, multiple login sessions, oversized file uploads, network loss, blank submissions, and unauthorized URL access.

EXP NO: 15

INVENTORY MANAGEMENT SYSTEM – TEST PLAN

Objective:

To develop a comprehensive test plan for an Inventory Management System, outlining the testing scope, design techniques, and execution strategies to ensure accurate tracking of stock, sales, and orders.

Test Plan

Test Approach: Manual Approach
Test Environment: Browser / Mobile Browser / Server
Training Needed: NO
Risk: NO

1. Admin Login

- Feature: Authentication
- Approach: Manual
- Test Data: Admin ID, Password
- Test Environment: Browser
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Open the Inventory Management System login URL.

Step 2 – Enter the valid Admin ID and password.

Step 3 – Click the Login button and verify that the dashboard appears.

Expected Result: The admin should be able to log in successfully and access the dashboard.

Status: Pass

2. Add New Product

- Feature: Inventory Entry
- Approach: Manual
- Test Data: Product Name, SKU, Quantity
- Test Environment: Browser
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Click the Add Product option.

Step 2 – Enter the product details, including name, SKU, quantity, and price.

Step 3 – Click Save and verify that the product is listed in the inventory.

Expected Result: The system should successfully add the new product and display it in the inventory list.

Status: Pass

3. Stock Update

- Feature: Stock Level
- Approach: Manual
- Test Data: Numeric Values
- Test Environment: Browser
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Select an existing product.

Step 2 – Increase or decrease the product quantity.

Step 3 – Verify that the total stock quantity is updated correctly.

Expected Result: The system should correctly update the stock level based on the quantity added or removed.

Status: Pass

4. Supplier Management

- Feature: Supplier Information
- Approach: Manual
- Test Data: Supplier Name, Contact Details
- Test Environment: Browser
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Add a new supplier with valid details.

Step 2 – Link the supplier to a product.

Step 3 – Save the information and verify that the supplier appears in the supplier list.

Expected Result: The system should successfully save supplier information and link the supplier to the selected product.

Status: Pass

5. Purchase Order

- Feature: Ordering
- Approach: Manual
- Test Data: Order ID, Supplier ID
- Test Environment: Browser
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Create a new purchase order.

Step 2 – Add the required products and quantities to the order.

Step 3 – Verify that the order status changes to "Sent".

Expected Result: The system should successfully create the purchase order and update its status to "Sent".

Status: Pass

6. Sales Tracking

- Feature: Transaction
- Approach: Manual
- Test Data: Sale ID, Customer Information
- Test Environment: Browser
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Record a new sale by entering the required product and customer details.

Step 2 – Verify that the stock quantity is reduced automatically after the sale.

Step 3 – Verify that the sales receipt is generated.

Expected Result: The system should record the sale, automatically update the stock quantity, and generate a receipt.

Status: Pass

7. Low Stock Alerts

- Feature: Notification
- Approach: Manual
- Test Data: Threshold Value
- Test Environment: Browser
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Set the low-stock alert threshold at 5 units.

Step 2 – Reduce the product stock quantity to 4 units.

Step 3 – Verify that an alert message appears on the dashboard.

Expected Result: The system should automatically generate a low-stock notification when the product quantity falls below the specified threshold.

Status: Pass

8. Barcode Scanning

- Feature: Product ID
- Approach: Manual
- Test Data: Barcode Serial Number
- Test Environment: Mobile / Browser
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Scan the barcode of a product.

Step 2 – Verify that the system identifies the product using its barcode.

Step 3 – Verify that the corresponding product details are displayed.

Expected Result: The system should identify the product correctly and display its details after scanning the barcode.

Status: Pass

9. Report Generation

- Feature: Analytics
- Approach: Manual
- Test Data: Monthly Date Range
- Test Environment: Browser
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Select the Monthly Sales Report option.

Step 2 – Choose the required date range, such as February 2026.

Step 3 – Generate the report and export it as a PDF or Excel file.

Expected Result: The system should generate the monthly sales report correctly and allow the admin to export it in PDF or Excel format.

Status: Pass

10. Database Backup

- Feature: Data Security
- Approach: Manual
- Test Data: SQL Backup File
- Test Environment: Server
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Click the Initiate Backup option.

Step 2 – Download the generated database backup file.

Step 3 – Verify the backup file and check the integrity of the stored data.

Expected Result: The system should successfully create and download the database backup while maintaining data integrity.

Status: Pass

Result:

Thus, a comprehensive test plan for the Inventory Management System was prepared by identifying its major features and defining the testing approach, test data, test environment, and test cases to verify admin login, product addition, stock updates, supplier management, purchase orders, sales tracking, low-stock alerts, barcode scanning, report generation, and database backup.